

EDA Asks What the Data Can Support

Exploratory data analysis (EDA) is the process of examining a dataset to understand its structure, quality, distributions, and relationships.



CORE CONCEPT

Start with Schema, Grain, and Quality Rules

A schema describes column names, data types, and expected meanings. A quality rule states what a valid value or row must satisfy. Examples: Data types describe storage. They do not guarantee meaning. A numeric age column can still contain impossible values.

age must be between 18 and 100;

customer ID must be unique;

balance may be negative, so "negative" is not automatically invalid;

"unknown" may be a missing-value marker rather than a real category.

RUN IT AND INSPECT THE RESULT

Center Describes a Typical Value

```
values = pd.Series([20, 22, 24, 26, 200])  
print(values.mean(), values.median())
```

OUTPUT

58.4 24.0

The value 200 pulls the mean far above most observations.

REASONING SEQUENCE

Spread Describes How Values Vary

Frame

Range: maximum
minus minimum

1

Transform

Variance: average
squared distance
from the mean

2

Validate

Standard deviation:
square root of
variance

3

Explain

Interquartile range (IQR):
third quartile minus first
quartile

4

For [2, 4, 6], mean is 4. Squared distances: $(2-4)^2 = 4$, $(4-4)^2 = 0$, $(6-4)^2 = 4$ Population variance: $(4 + 0 + 4) / 3 = 2.67$ Population standard deviation: $\sqrt{2.67} \approx 1.63$

CORE CONCEPT

A Distribution Shows Frequency and Shape

A distribution describes which values occur and how often. A histogram divides a numerical range into intervals called bins. Bar height shows how many observations fall inside each interval. Common shapes include:

roughly symmetric;

right-skewed with a long high-value tail;

left-skewed with a long low-value tail;

multimodal with more than one peak.

RUN IT AND INSPECT THE RESULT

Outliers Are Signals to Investigate

```
q1 = df["value"].quantile(0.25)
q3 = df["value"].quantile(0.75)
iqr = q3 - q1
outliers = df[~df["value"].between(q1 - 1.5*iqr, q3 + 1.5*iqr)]
```

OUTPUT

A DataFrame containing values outside the stated fences.

A value of 95 is flagged because $95 > 70$.

CORE CONCEPT

Missing Markers and Categories Need Counting

Missing information may appear as NaN, an empty string, -1, "unknown", or "not applicable". These are sentinel values when they stand in for another meaning. For categorical columns: A rare category may be valid and important. Combining categories requires a stated reason.

count each category;

include missing values;

inspect spelling and capitalization;

confirm whether sentinel values are real categories;

report both counts and percentages.

REASONING SEQUENCE

Class Imbalance Changes How Accuracy Feels

Frame

A target is the outcome a supervised model will learn to predict.

1

Transform

For a categorical target, each possible value is a class.

2

Validate

Class imbalance occurs when one class is much more common than another.

3

Explain

If 90 of 100 customers did not subscribe, a rule that always predicts "no" is 90% accurate.

4

Actual counts: Always predict No: accuracy = 90 correct / 100 total = 0.90
Subscribers found: 0 / 10 = 0.

RUN IT AND INSPECT THE RESULT

Relationships Do Not Automatically Mean Causation

```
print(df[["study_hours", "score"]].corr())
```

OUTPUT

A 2-by-2 correlation matrix with 1.0 on the diagonal.

Suppose study hours and scores have correlation 0.70. This supports a positive association in this dataset. It does not prove that adding one study hour causes a fixed score increase.

QUALITY GATE

Feature Engineering Creates Measurable Inputs



contactdurationminutes = seconds / 60



haspreviouscontact = previouscontacts > 0



extract month from a valid contact date



a clear meaning;

QUALITY GATE

A Reproducible EDA Has a Question and Evidence



state the question;



identify required columns and row grain;



check missing, invalid, and duplicate values;



calculate a relevant statistic;

GUIDED LAB

Guided Lab: Produce Three Defensible Findings

01

**row grain and
target definition**

02

**shape, types, missing
markers, and
duplicate checks**

03

**target class
counts and
percentages**

04

**one numerical
distribution with
center and spread**